Environment and data	Still images	Video memory	Live runtime	Multi-camera
Phase 0	Phase 1	Phase 2	Phase 3	Phase 4
DONE	DONE	DONE, MIXED RESULT	DONE, BELOW 30 FPS	DONE, RGB ONLY
Kaggle 2×T4, laptop RTX 4050	two YOLO26-n backbones	Temporal State Memory	capture thread per camera	RGB YOLO26-m, seven views
LLVIP, M3FD in a paired layout	CMFM reliability-gated scan	trained on fault-injected clips	stream monitor, sensor flags	ground-plane projection, merge
KAIST, WILDTRACK added later	baselines, ablations, faults	ByteTrack on KAIST sequences	half-precision loop, tracker	ground tracks and counts
Reached: one fixed M3FD split for every run.	*Reached:* fusion beats best single camera; scan not significant over concat.	*Reached:* fewer misses in thermal noise (confounded); no tracking gain.	*Reached:* flat delay for ten minutes; below 30 fps; flags give no gain.	*Reached:* counts and tracks on RGB only; no evidence on fusion.

Model across the phases

All blocks are built and measured; multi-camera ran on its own RGB detector.

